

INFOGRAPHIC: Bayesian Methods

Updating Beliefs in Statistical Modelling

[FREQUENTIST]

Parameters (θ) are **fixed, unknown constants**.
Probability is the long-run relative frequency of data.

[BAYESIAN]

Parameters (θ) are **random variables** with probability distributions. Probability represents a degree of belief.

1. The Philosophical Shift

In Bayesian statistics, we don't just ask "what do the data say?". We ask: *"How should this new data update what I already believed?"*

- **Prior Knowledge:** Seamlessly integrates historical data, expert opinions, or scientific constraints before seeing the current data.
- **Intuitive Interpretation:** A 95% Bayesian Credible Interval means there is a 95% probability that the parameter lies within that interval (unlike Frequentist confidence intervals).

2. Bayes' Theorem for Statistical Modelling

The foundation of all Bayesian inference maps prior beliefs into updated knowledge via the likelihood function.

$$p(\theta | X) = [p(X | \theta) \times p(\theta)] / p(X)$$

● **Posterior Distribution** — $p(\theta | X)$: Your updated belief about the parameter θ after looking at the observed data X .

● **Likelihood Function** — $p(X | \theta)$: How likely the observed data is under different parameter values (the data's signal).

● **Prior Distribution** — $p(\theta)$: Your initial uncertainty/belief about θ before collecting any data.

● **Marginal Likelihood (Evidence)** — $p(X)$: A normalizing constant ensuring the posterior integrates to 1. Computed as $\int p(X | \theta)p(\theta)d\theta$.

In short: **Posterior** $\propto$ **Likelihood** $\times$ **Prior**

3. The Conjugate Prior vs. MCMC Reality

Conjugate Priors (The Analytical Route)

When the Prior and the Posterior belong to the same probability family, the prior is called *conjugate*. This allows exact mathematical solutions without computational simulation.

- **Beta Prior + Binomial Data = Beta Posterior** (Classic example for proportions)
- **Normal Prior + Normal Data = Normal Posterior** (Classic example for means)

Markov Chain Monte Carlo - MCMC (The Modern Route)

For complex high-dimensional models, calculating the Evidence $p(\mathbf{X})$ analytically is impossible. Instead, we use computational simulation algorithms (like **Metropolis-Hastings** or **Gibbs Sampling**) to draw random samples directly from the unnormalized posterior distribution.

4. Bayesian Estimation Steps

1. Define a probability model for the data given parameters (the **Likelihood**).
2. Formulate a probability distribution reflecting your initial state of knowledge (the **Prior**).
3. Collect data and compute the updated parameter distribution (the **Posterior**).
4. Analyze the posterior metrics (e.g., Posterior Mean, Credible Intervals) to make inferences.